Name _____ _____ Baggett/Fall2026/207/Quiz2

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 2 for the reason.**

1. (**3** pts) $\lim_{x\to 1} \dfrac{x-3}{x^2+2x-4} = \dfrac{\lim_{x\to 1}(x-3)}{\lim_{x\to 1}(x^2+2x-4)}$

True **False**

Why? _____ /3

2. (**3** pts) If $f(0.1)$, $f(0.01)$ and $f(0.001)$ all round to 2, then $\lim_{x\to 0} f(x) = 2$.

True **False**

Why? _____ /3

3. (**5** pts) Let

$$g(x) = \begin{cases} 2x+1, & x < 1 \\ 5, & x = 1 \\ 6-3x, & x > 1 \end{cases}$$

(a) $\lim_{x\to 1^-} g(x) =$ _____ (b) $\lim_{x\to 1^+} g(x) =$ _____

(c) $\lim_{x\to 1} g(x) =$ _____ (d) $g(1) =$ _____

(e) Is g continuous at $x = 1$? If not, say which requirement fails.

_____

_____ /5

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4. (**4** pts) Evaluate the limit. Show the rewriting — substituting first and finding $\frac{0}{0}$ is not an answer, it is the reason a rewrite is needed.

$$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$$

/4

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5. (**3** pts) Let $f(x) = \dfrac{x+1}{x-3}$. **No calculator.** Decide each limit by naming the sign and size of the numerator and of the denominator.

(a) $\lim\limits_{x\to 3^-} f(x) =$ _____ (b) $\lim\limits_{x\to 3^+} f(x) =$ _____

(c) Write the equation of the vertical asymptote.

/3

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6. (**2** pts) Find the value of the constant c that makes

$$h(x) = \begin{cases} x^2 - 1, & x < 3 \\ cx + 2, & x \geq 3 \end{cases}$$

continuous at $x = 3$.

/2

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